

CAIE IGCSE Chemistry

2.3 Isotopes

Notes

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Define isotopes as different atoms of the same element that have the same number of protons but different numbers of neutrons

- Isotopes are atoms of the same element that have the same number of protons and electrons but different number of neutrons
 - E.g. Carbon-12 has 6 protons and 6 neutrons but carbon-13 has 6 protons and 7 neutrons in the nucleus of each atom.

Interpret and use symbols for atoms, e.g. $^{12}_6\text{C}$, and ions, e.g. $^{35}_{17}\text{Cl}^-$

- In a periodic table, the elements are written with their mass number on the top and their atomic number on the bottom
- The atomic number can be used to work out the mass number and vice versa
 - E.g. $^{12}_6\text{C}$ has the atomic number 6 so has 6 protons, so the mass number is 12 (6 protons and 6 neutrons)
- For ions, the atomic number and the mass number stays the same as the original atom, but a minus or plus symbol is added after the letters to indicate whether the ion is positive or negative
 - E.g. $^{23}_{11}\text{Na}^+$ has the mass number 23 and atomic number 11, but the plus symbol shows that an electron has been lost, so the overall charge of the ion is +1.
- The atomic number can be used to work out the mass number and vice versa, and the charge can be used to work out the number of electrons:
 - E.g. $^{35}_{17}\text{Cl}^-$
 - The atomic number shows that the chloride ion has 17 protons
 - So if the mass number is 35, there must be 18 neutrons (17 protons + 18 neutrons = 35)
 - The minus symbol shows that an electron has been gained and the overall charge of the ion is -1, so there must be 18 electrons (17 electrons + 1 gained electron)



(Extended only) State that isotopes of the same element have the same chemical properties because they have the same number of electrons and therefore the same electronic configuration

- Isotopes of the same element will have the same chemical properties because the number of electrons stays the same, so the electron configuration stays unchanged

(Extended only) Calculate the relative atomic mass of an element from the relative masses and abundances of its isotopes

- The relative atomic mass is an average value that takes account of the abundance of the isotopes of the element
 - Example question: Carbon has 2 isotopes= Carbon-14 with abundance 20% and carbon-12 with abundance 80%. Calculate the relative atomic mass of carbon.
 - Equation to work out relative atomic mass:

$$\frac{(\text{isotope 1 mass} \times \text{abundance}) + (\text{isotope 2 mass} \times \text{abundance})}{100}$$

- For this question: $((14 \times 20) + (12 \times 80)) \div 100 = 1240 \div 100 = 12.4$

